Phonologically conditioned affix order is stratal

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In this paper, we identify a previously unnoticed prediction of Stratal OT that affixes may be reordered for phonological optimization, and argue that the prediction is in fact borne out.

Keywords: StratalOT, affix order, cyclicity

Spark

In this paper, we identify a previously unnoticed consequence of Stratal Optimality Theory (StratOT) (Kiparsky 2000, 2015; Bermúdez-Otero 2011, 2016): Phonologically conditioned affix order (henceforth: pcao) applies strictly within one stratum. Whereas Paster (2006, 2009) dismisses PCAO as typologically unattested and argues that apparent instances of this phenomenon should be analyzed in other ways, Jenks & Rose (2015) show that a pcao-analysis for the Moro and Huave patterns discussed below is clearly superior to the alternative interpretations suggested by Paster. The same point is made by Benz (2017) for Washo, and similar evidence can be found in Rarámuri Caballero (2008) and Yidiɲ (Popp 2024). Here, we want to argue that pcao also provides evidence for the stratal architecture of phonological grammars.

In contrast to Parallel OT (Prince & Smolensky 1993), StratOT assumes multiple levels (*strata*), in each of which the morphological component applies prior to the phonological component.[1] In (1), we provide a toy example which illustrates both the general architecture of StratOT and its application to PCAO. At the first stratum, the *stem level*, the affixes -*o* and -*mat* attach to the root *le*, which produces the input to the phonological component of the first stratum /*le-o-mat*/. In the phonology of the stem level, the grammar still has access to morpheme boundaries (indicated by '-' in (1)). As a consequence, reordering the affixes is a natural strategy to reduce phonological markedness, alongside other phonological processes: -*o* and -*mat* switch their positions to avoid a hiatus in /*le-o-mat*/ in (1). After the first stratum, *Bracket Erasure* (Pesetsky 1979; Kiparsky 1982) applies, making morphological boundaries inaccessible to further operations. At the next stratum, the *word level*, the affix -*on* is attached. After this step, the phonology of the word level receives the input /*lemato-on*/, which contains a hiatus, just as /*le-o-mat*/. Due to Bracket Erasure after the stem level, however, the phonology of the word level is blind to the morpheme boundaries of the previous stratum. Consequently, reordering of the affixes to reduce markedness is not possible (**le-mat-on-o*).

(1)

morphology	*le* + *o* + *mat*	stem level
phonology	/*le-o-mat*/ → [*le-mat-o*]	
Bracket Erasure		
morphology	*lemato* + *on*	word level
phonology	/*lemato-on*/ → [*lemato-ʔon*], *[*lemat-on-o*]	
Bracket Erasure		

This model makes the strong prediction that only affixes of the same stratum can be reordered for phonological reasons. Crucially, affixes of a later stratum cannot participate

[1] Here we assume a modular architecture where PCAO is derived in a phonological evaluation following and independently from morphology proper. However, the same predictions for stratal locality would also follow in a system where phonological and morpheme-specific ALIGNMENT constraints apply simultaneously as assumed in the analysis of Benz (2017) for Washo.

since they are not part of the word at the point where pcao applies. Affixes of a previous stratum cannot be reordered because Bracket Erasure has erased their morpheme boundaries, indispensable information for pcao to apply. Hence, reordering of stem level affixes is not a possible option for the phonology of the word level, thus contrasting the phenomenon with phonological metathesis, which might still affect phonological material of earlier levels.

In the literature, there are two cases of pcao which have been explicitly described as intra-stratal optimization. Caballero (2010) argues that the evidential marker /*-ča(ne)*/ in **Choguita Rarámuri** (Uto-Aztecan, Mexico) requires a trochaic foot on its left. To achieve this, it may swap position with other suffixes (but may not infix into roots). Crucially, this is only possible within the stratal domain that Caballero (2010) refers to as aspectual stem. Benz (2017) provides a detailed analysis of pcao in **Washo** (isolate, USA). In her account, the position of underlyingly stressed affixes avoids phonologically marked patterns, such as sequences of two stressed syllables. As Benz (2017) shows, only suffixes of the same stratum are involved in pcao.

The bounding of reordering to a single stratal domain also holds for the best-known case of pcao, **San Francisco del Mar Huave** (Mexico, Kim 2008, 2010), where the position of certain monoconsonantal affixes to the verb depends on phonological properties, yet their relative order with respect to other affixes does not change. For example, the 1st person subordinate marker *n* appears as a suffix in (2-a) but a prefix in (2-b). Kim (2008, 2010) argues that these affixes are suffixes by default but shift to prefixal positions with vowel-initial stems, such as *ajch* in (2-a), to optimize the phonological structure. Reordering takes place to prevent the application of vowel epenthesis, which resolves consonant clusters otherwise, as seen in (2-b). Kim demonstrates that this reordering can be implemented as cyclic phonological evaluation, but in a stratal architecture, application of a process P in a given cycle a fortiori implies that P applies also in a single stratum since cycles are properly contained in strata. Thus Huave fully fulfils our hypothesis.

In fact, there is independent evidence that affix reordering in Huave is closely connected to stratal organization. Of the 4 affixal layers Kim invokes to informally describe Huave affix order, only the three inner ones exhibit affix mobility. In the outermost affix layer, suffixes are not reordered under phonotactic pressure. Thus the plural suffix *-n* which is homophonous to the 1st person subordinate, but part of the outer layer does not move to a prefixal position but remains in suffix position where it triggers vowel epenthesis (2-c). A natural interpretation in terms of strata is that affix dislocation in Huave is only possible at the Stem Level (corresponding to the three inner layers), but not at the Word Level (the outermost layer) due to the different ranking of markedness constraints. This obviates the necessity to posit morpheme-specific constraints to capture the different behavior of plural and subordinate *n* in Kim's analysis. This interpretation is also in line with the detailed argument for a stratal overall organization of the closely related variety of San Mateo del Mar Huave supported by Noyer (2013), where he assigns the cognate plural suffix to the Word Level (in Noyer terms, the 'noncyclic phonology') based on segmental and prosodic differences.

(2) Mobile affix placement in Huave (Kim 2008:153, 324, 333)

a.	x-i-$\boxed{\text{n}}$-ajch	b.	pajk-a-$\boxed{\text{n}}$	c.	i-m-e-jch-V-$\boxed{\text{n}}$
	1-FUT-1SUB-give		face.up-V-1SUB		FUT-SUB-2-give-V-PL
	'I will give'		'(that) I lie face up'		'you will give'

In **Yidiɲ** (Pama-Nyungan, Australia), some suffixes lengthen the preceding syllable (indicated by a ⁺), like the andative suffix -⁺*li*. In combination with monosyllabic suffixes, its position depends on the syllable number of the root. With the bisyllabic root *magi* in (3-a), the AND suffix follows the root. With the trisyllabic root *maɖinda* in (3-b), the AND suffix follows the COMIT suffix -ŋ*a*. Dixon (1977) concludes that the AND suffix appears in those positions where it lengthens an even-numbered syllable (the 2nd syllable in (3-a), the 4th syllable in (3-b)). The INTRANS suffix ⁺*ɖi* in (3-c) also lengthens its preceding vowel. In (3-d), it combines with a trisyllabic root and a monosyllabic, inflectional PURP suffix *-na* creating a phonologically similar context to (3-b). In (3-d), however, affix reordering does not take place (see (3-e)). This reveals an asymmetry between derivational and inflectional suffixes: derivational suffixes participate in PCAO, inflectional suffixes do not despite having the necessary phonological shape. This asymmetry follows automatically from the assumption that inflectional suffixes attach at a later stratum, thus counter-feeding PCAO. In Yidiɲ, there is independent evidence for stratification from a different process called final syllable shortening. In this process, derivational and inflectional affixes show non-uniform behaviour suggesting that the two affix groups are in fact affected by different phonological grammars, as shown in Popp (2024).

(3) Phonologically conditioned affix order in Yidiɲ (Dixon 1977:228)

a. /magi-$\boxed{^{+}\text{li}}$-ŋa-l-ŋ/
[(ma.giː)(ɻi.ŋal)]
climb.up-AND-COMIT-CL-PRES
'going to climb up with sb.'

b. maɖinda-ŋa-$\boxed{^{+}\text{li}}$-ŋ
[(ma.ɖin)(da.ŋaː)liŋ]
walk.up-COMIT-AND-PRES
'going to walk.up with sb.'

c. ɖara-$\boxed{^{+}\text{ɖi}}$-ŋada-ŋ
[(ɖa.raː).(ɖi.ŋa)daŋ]
stand-INTRANS-VEN-PRES
'come and stand'

d. wuŋaba-$\boxed{^{+}\text{ɖi}}$-na
[(wu.ŋa).(ba.ɖiː).na]
hunt.for-INTRANS-PURP
'hunt for'

e. *[(wu.ŋa).(ba.naː).ɖi]

We are aware of only one example where phonologically conditioned affix order has been claimed to straddle across stratal domains, Jenks & Rose (2015) in their analysis of **Moro** (Kordofanian, Sudan) object markers. These are suffixal in most contexts, but in paradigms showing default tone (a single H with local spreading), H-toned object markers appear in prefix position without a further H-tone on the root. We follow Jenks & Rose in understanding this as shifting of an object marker to provide a H-tone at the left edge of a domain they call MACROSTEM: A high-ranked Align constraint requires a H-tone at the left edge of this stem domain, which is either satisfied by shifting a H-toned object suffix or by inserting a default H on the root (Moro has no lexical tone contrast on verb roots). Jenks & Rose (2015) argue that the dislocation applies across three different domains. However, the two most relevant domains are only evidenced by one phonological process. For each of these processes, there are alternative explanations for

their locally restricted applications. For example, Jenks & Rose adduce vowel harmony as evidence for a domain boundary where the ATR-value of the root spreads iteratively to the left, but only locally to the TAM-vowel on the right. However, it is well-known that harmony systems show directional asymmetries of this type for purely phonological reasons. Thus Chilcotcin has a similar symmetry (iterative spreading to the left and non-iterative spreading to the right) for RTR which does not correspond to specific morphological constituents (see Mullin 2011 for detailed discussion and analysis).
Let us finally consider the bigger context of affix ordering in stratal models. It has been a central debate in Lexical Phonology whether stratal affiliation at least partially determines affix order or not. We take here the stance of Kiparsky (2020) that the major source of apparent violations of the Affix Ordering generalization are dual-level affixes whose stratal affiliation is contingent on independent morphological factors. However, this still implies that for morphological reasons the ordering of a single affix *may* straddle the boundaries of different strata in contrast to what we have suggested for PCAO. On the other hand, there are affix ordering patterns which are generally believed to be morphological in nature, but still seem to scrupulously respect boundaries of independently motivated strata. A case in point is the Bantu CARP template whose domain has been shown to systematically coincide with verbal stems by Good (2016). And of course there are cases of stratally bound affix reordering for which it is still unclear whether they are phonologically or morphologically triggered (see, e.g., Buckley 1994 on Kashaya). It is also conceptually far from settled which types of pcao are derived in the phonology. Whereas phonological subcategorization (determining for example PCAO in the Rarámuri case cited above) is assumed to be evaluated in phonology proper by Inkelas (1990) and Bermúdez-Otero (2018), other authors interpret it as a mechanism strictly in the morphology (e.g., Paster 2006, 2009; Kalin 2020). On this background, the asymmetry in phrasal bounding predicted by our hypothesis might turn out to be an important diagnostic for distinguishing morphologically and phonologically conditioned affix order.

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